DANDAN SANG (LUNA)

West Vancouver, BC
(236) 866-6533 ◇ Luna_sang@sfu.ca

EDUCATION

Master of Science in Computer Science 05/2026 – Present
Simon Fraser University, Burnaby, BC, Canada

Master of Science in Big Data 09/2025 – 04/2026
Simon Fraser University, Burnaby, BC, Canada

Bachelor of Science in Computer Science 01/2022 – 09/2025
Simon Fraser University, Burnaby, BC, Canada

Master of Science in Finance 09/2013 – 09/2014
Imperial College London, UK

Bachelor of Science in Business Finance and Economics 09/2010 – 06/2013
University of East Anglia, Norwich, UK

RESEARCH INTERESTS

Safe Learning, Deep Learning, Control Theory, Learning-based Control, Hamilton–Jacobi Reachability, High-dimensional Reachability, Trustworthy AI

EXPERIENCE

Research Assistant 2024 – June 2026
MARS Lab, Simon Fraser University *Safe Learning in the Real World via Adaptive Shielding*

- Trained and evaluated robots using a robust shielding framework for safe reinforcement learning in real-world scenarios.
- Contributed to the research and publication of the paper "Safe Learning in the Real World via Adaptive Shielding with Hamilton–Jacobi Reachability," published at Learning for Dynamics and Control (L4DC 2025).

Research Assistant Fall 2024
Rosie Lab, Simon Fraser University *Exploration and Visualization of BERSt Dataset*

- Contributed to the BERSt project, focusing on the analysis of prosodically emotional speech from professional voice actors.
- Performed feature extraction using OpenSMILE and visualized vocal features by emotion, affect, and distance.
- Co-authored the paper "BERSting at the Screams: A Benchmark for Distanced, Emotional, and Shouted Speech Recognition," published in *Computer Speech & Language* (2025).

Research Assistant (USRA) Summer 2024
MARS Lab, Simon Fraser University *Robotic Follow-Ahead*

- Contributed to the Robotic Follow-Ahead project by assisting in model training, data preparation, and testing protocols.
- Preprocessed and transformed the SFU nav store dataset, implementing interpolation and resampling techniques.
- Developed a testing application integrating sensors for human-activity tracking and optimized system accuracy.

TECHNICAL PROJECTS

Predict Students' Dropout and Academic Success Sep 2024 – Dec 2024
Data Mining, SFU

- Preprocessed data including missing values, encoding, and PCA, and performed EDA to identify key patterns.
- Trained and evaluated models including Random Forest, SVM, and k-NN, achieving high F1 and ROC-AUC.

Emotion-Aware Chatbot with Real-time Emotion Recognition Jan 2024 – May 2024
Affective Computing, SFU

- Built an emotion-aware chatbot integrating multimodal inputs via facial expression recognition.
- Used DeepFace, Hsemotion, and GPT-4V (Vision) for emotion interpretation and adaptive response generation.

Hotel Review Data Analysis – Big Data & ML May 2023 – Aug 2023
Computational Data Science, SFU

- Applied NLP techniques for sentiment analysis on New York hotel reviews using Spark, Hadoop, and scikit-learn.

GLIMPS – GitLab Analyzer May 2023 – Aug 2023
Special Project, SFU

- Enhanced a professor-used tool to analyze GitHub/GitLab contributions, migrating infrastructure to a GitHub server.

OTHER WORK EXPERIENCE

Senior Associate Oct 2016 – Jan 2021
Deloitte LLP, Beijing, China

- Led multiple financial audit engagements for large-scale clients, including mining and manufacturing companies.
- Designed audit procedures for M&A and related-party transactions, ensuring compliance and accuracy.

Associate Jan 2015 – Sep 2016
Deloitte LLP, Beijing, China

PUBLICATIONS

- Paige Irene Tuttosi, Mantaj Dhillon, Luna Sang, Shane Eastwood, Poorvi Bhatia, Quang Minh Dinh, Avni Kapoor, Yewon Jin, and Angelica Lim. "BERSting at the Screams: A Benchmark for Distanced, Emotional, and Shouted Speech Recognition." *Computer Speech & Language*, 2025.
- Michael Lu, Jashanraj Gosain, Luna Sang, and Mo Chen. "Safe Learning in the Real World via Adaptive Shielding with Hamilton–Jacobi Reachability." *Learning for Dynamics and Control (L4DC)*, 2025.

SKILLS

Programming	Python, C++, C, Java, JavaScript, TypeScript, and SQL
Machine Learning	PyTorch, scikit-learn, NumPy, SciPy, pandas
Robotics & Simulation	ROS, ROS 2, Gazebo, robot learning, sensor integration, and simulation
Computing & Data	Spark, Hadoop, GPU computing, Slurm, and Linux
Development Tools	Git, GitHub, VS Code, Jupyter, and Overleaf